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ML-Based Detection of ARP Spoofing Attacks in Internet-of-Things Environments: Comparative Study

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Abstract

The Address Resolution Protocol (ARP) spoofing is a common method used by adversaries to attack the security of Internet of Things (IoT) networks through Man-in-the-Middle (MITM) attacks. Lightweight, resource-constrained IoT devices are often vulnerable to these attacks; thus, traditional detection mechanisms are often insufficient to identify and counteract them in real time. In this study, we proposed a machine learning-based model to detect ARP spoofing MITM attacks using the IoTID20 dataset. The detection model is systematic and includes dataset extraction, preprocessing, train-test splitting, classifier training, and performance evaluation. We used accuracy, precision, recall, and F1-score to evaluate eleven machine learning classifiers, including tree-based ensembles, linear models, kernel-based methods, and optimization-driven algorithms. Results show that ensemble methods performed the best, with LightGBM achieving the highest score of 99.7% across all four metrics, followed by XGBoost and Random Forest. Simpler algorithms like logistic regression and ridge performed less well but still delivered competitive results, indicating their potential for deployment on resource-limited IoT devices. These findings demonstrate that machine learning can effectively detect ARP spoofing attacks with high accuracy and efficiency, offering a promising approach to securing IoT environments against MITM threats.

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1. Introduction

Internet of Things (IoT) has been quickly adopted as a paradigmatic shift, bringing billions of heterogeneous IoT devices into the healthcare field, transportation, industrial automation, and smart homes [1, 2]. Although this connection, being widespread, opens opportunities, it also opens up IoT networks to serious security vulnerabilities because of their resource-constrained design, the absence of standardized security measures [5-3], and the use of lightweight communication protocols. Address Resolution Protocol (ARP) spoofing attack is one of the most dangerous attacks in this context and is widely used to provoke Man-in-the-Middle (MITM) attacks [6]. With the help of manipulating ARP tables, opponents may intercept, modify or prevent communication between devices, thus violating data confidentiality, integrity, and availability [7, 8]. Conventional security systems like firewalls and intrusion detection systems do not usually work well in the IoT setting [9], because they demand significant computing capabilities and are not programmed to identify protocol-specific attacks, such as ARP spoofing. Additionally, signature-based methods are unable to identify new or advanced variations of attacks, which exposes IoT infrastructures to attack. Such constraints will require the establishment of smart and self-developing detection systems.

Recent progress in machine learning (ML) [10] has yielded potential in overcoming network intrusion detection problems and provides the capability to learn regularities of malicious activity and generalize to zero-day attacks [11] [12, 13]. ML-based detection systems can profile traffic, distill discriminative attributes, and construct classifiers that have the capability of distinguishing malicious and benign packets [14, 15]. Nevertheless, despite the usage of ML in more general intrusion detection problems in the IoT networks, a scientific gap remains in developing specific ML-based systems in identifying ARP spoofing MITM attacks within the context of the IoT [16, 17].

ARP spoofing in the IoT networks does not only cause data theft. Effective MITM attacks may enable massive Distributed Denial of Service (DDoS), bring malicious code to the communication of devices, and disrupt vital IoT services like health devices and industrial control systems [18, 19]. Research has indicated that in such cases, the security breach of IoT devices not only interferes with the effectiveness of the devices but might also lead to financial losses and risks to human lives. This increases the necessity of the development of lightweight but powerful detection solutions that could be incorporated into the IoT infrastructures.

The heterogeneity of IoT devices, the dynamic nature of network traffic, and the scarcity of labeled datasets further complicate the issue of detecting ARP spoofing in IoT environments. This necessitates the need to scout hybrid methods of integrating data preprocessing, feature engineering [20-22], and optimized ML classifiers [23]. This study relies on recent experiments that focus on the effectiveness of ensemble learning, deep learning models, and hybrid optimization techniques in enhancing the accuracy of detection and ensuring low computational costs [24, 25].

To address this gap, this paper puts forward an ML-based detection system of ARP spoofing MITM attacks in the IoT. The offered solution uses the characteristics of network traffic to train the models of classification that differentiate between the legal communication and the ARP packets containing the spoofing. Our research adds value by testing the performance of various ML classifiers to detect ARP spoofing MITM attacks and determine the best ML classifiers for detecting ARP spoofing MITM attacks in IoT.

The remaining part of the paper is structured in the following way: Section II is the related work review, ARP spoofing detection, and ML-based IoT security. Section III explains the suggested methodology, such as data collection, feature extraction, and model design. Section IV shows the experimental findings and performance analysis. In section V, there are findings and implications. Lastly, the paper ends with Section VI, which summarizes the future research direction.

2. Related works

The IoT security has become a highly researched area over the past few years due to the explosive growth of connected devices and the increase in the number of network-based attacks [26, 27]. There is a lot of literature on intrusion detection systems designed to protect IoT networks, both traditional rule-based and signature-based systems, and more flexible anomaly detection systems [28, 29]. Specifically, ARP spoofing and (MITM) attacks have attracted quite a significant treatment because of their capacity to use protocol-level vulnerabilities and breach device-to-device communication. Although the initial approach to detection was based on the detection of ARP table inconsistencies

and the implementation of cryptographic mechanisms, they either placed a significant computational load on a computer or did not align with their changing patterns of an attack. This has motivated scientists to appeal more to ML-based methods, which offer more flexibility, scalability, and the ability to identify known and new variations of ARP spoofing in IoT systems [30, 31]. This part represents the latest related works.

A study by Usmani et al. [32] has introduced an approach based on machine learning to predict ARP spoofing attacks in IoT networks on the Kitsune dataset that contains a variety of man-in-the-middle (MITM) attacks. Two models, Long Short-Term Memory (LSTM) networks and Decision Trees, were evaluated in their study to classify malicious and benign packets. The findings revealed that both models produced incredibly high accuracy, with LSTM producing up to 99.9% and Decision Tree producing 100 %. Interestingly, the Decision Tree was able to perform much faster than the LSTM, as well as point towards which features alone were good enough to reach near-perfect detection. The authors also considered a hybrid solution, where the features chosen by Decision Trees were sent to LSTM, leading to less training time and the loss of only a small part of accuracy. This paper shows that lightweight ML methods can be successfully used to identify ARP spoofing, and that a simpler classifier can be more useful in resource-constrained IoT applications than alternatives based on deep learning.

Another study by Prasad and Chandra [33] suggests a two-layered defense against ARP-spoofing-based MITM attacks, which is a machine-learning detector with a device-profiling module. Their ML pipeline ranks features with mutualinfoclassif and constructs a small set of features in a forward feature-selection process and classifies traffic with a VotingClassifier ensemble (AdaBoost, logistic regression, and k-NN) trained on the TON-IoT dataset. Their experiments showed very high performance of combined ML (accuracy 99.2%, precision 99.4%, recall 99.7%, F1 99.5%). To supplement ML, the device-profiling component implements a small client application on the endpoints, which monitors ARP cache entries and reports them to a centralized profiler; the profiler maintains a DEVPROFILE, detects duplicate/spoofed MAC entries, and blocks the intruder. and in their experiments, it has found that the profiling technique achieves 100% detection. The authors point out that having the client app installed on every device is an effective constraint of a large or highly dynamic network and recommend future work to eliminate this dependency.

Alsaaidah et al. [34] created an IoT-network machine learning-based model to detect ARP spoofing attacks, with a focus on a trade-off between the computational complexity and the detection accuracy. The research used the dataset of IoTID20, which deals with the ARP spoofing cases, and subjected a series of preprocessing tasks such as cleaning of features, normalization, and dimensionality reduction. Three feature selection methods, such as correlation filtering, wrapper-based forward selection, and Firefly inspired Optimization (FFO) [35], were evaluated with an extensive selection of classifiers, both simple (DT, KNN, SVM, NB) and ensemble-based (RF, XGBoost, CatBoost, AdaBoost, Gradient Boosting, Voting). It was found that Random Forest using wrapper-selected features scored the highest per cent accuracy (99.74) and a runtime of 305 ms, whereas Decision Tree using all features scored a lower per cent (99.41) but in only 11 ms, which is more appropriate in time-sensitive IoT applications. The authors determined that wrapper-based feature selection is a good compromise between accuracy and complexity, but the experiment was only conducted on the IoTID20 and recommends future research using newer databases.

Research by Kumar and Dash [36], real-time ARP spoofing detection and prevention based on machine learning algorithms were compared. They tested five models, namely: Random Forest (RF), Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), Support Vector Machines (SVM), and Isolation Forest on the captured ARP traffic to measure the accuracy, precision, recall, F1-score, and error rates. Findings showed that Random Forest performed optimally with the highest accuracy (94-95%), precision (92%), recall (95%), and F1-score (93%), which was the highest, hence it could be deployed in real time because false positives (5) and uncycled are low. Strong detection performance was also shown by LSTM and CNN, especially in terms of sensitising temporal and spatial characteristics of ARP traffic, but at a high computational cost and training duration. SVM had moderate accuracy (~88%), whereas Isolation Forest had the least detection reliability (~85%) and high error rates. The authors find that ensemble (such as Random Forest) approaches provide the best trade-off between detection and efficiency in the context of real-world IoT security, but note that problems such as diversity in data, computational costs, and adversarial evasion attacks are present.

A study by Vajrobol et al. [30] suggests a multiclass ML system that identifies both ARP and DNS spoofing in IoT systems using the CICIoT2023 dataset. The authors pre-process the data by SMOTE, label encoding, and Min-Max scaling, perform an 80:20 train-test split ([?]58,742 train / 14,686 test on the entire dataset), and test various classifiers: Decision Tree, Random Forest, XGBoost, a mixed model of Random Forest and XGBoost, and Gradient

Boosting. Random Forest performed the most overall (Accuracy = 95.1%, Precision = 95.2%, Recall = 95.1%, F1 = 95.1%), followed by XGBoost and the hybrid one, and then Decision Tree (~90.9%). The paper has demonstrated the strength of RF in this multiclass spoofing task (confusion matrices demonstrate good per-class detection), and has also indicated its weaknesses, such as class imbalance, the use of the CICIoT2023 lab dataset, and that additional real-world validation and investigation of deeper/hybrid models are required.

Another study by Sumanasekara and Abeysinghe [37] introduced an ML architecture based on behavior analysis in identifying ARP spoofing attacks in IoT networks. They designed behavioral characteristics (based on the CIC IoT 2023 dataset in PCAP format, filtered and converted into CSV), including ARP request frequency, inconsistencies between IP-MAC mappings, rolling MAC changes, unsolicited replies, and frequency deviations. They compared several classifiers (Linear SVC, SVC with RBF kernel, Logistic Regression, Gaussian Naive Bayes, KNN and an ANN with ReLU activation) after preprocessing (normalization, encoding, SMOTE balancing and feature selection with Random Forest importance). KNN was the most accurate, with equal precision, recall, and F1-scores (0.93-0.95), followed by ANN (82%), SVC (79%), and Logistic Regression (75%), and Naive Bayes was the slowest at 61%. The paper highlighted the importance of well-designed behavioral characteristics in enhancing ML to detect ARP spoofing patterns and suggested the extension of the framework with deep learning and hybrid models to work on it in the future.

Although the current literature has asserted machine learning as a promising tool to counter ARP spoofing attacks in the IoT context, a few issues persist. The accuracy of Usmani et al. [32] and Prasad and Chandra [33] is almost near perfect with the Kitsune and TON IoT datasets, but the solutions are dependent on individual datasets and might not apply to more heterogeneous IoT traffic. On the same note, Alsaaidah et al. [34] and Kumar and Dash [36] stressed the trade-off between accuracy and the cost of computation, in which deep learning models performed well, yet were impractical in resource-constrained devices. Work by Vajrobol et al. [30] and Sumanasekara and Abeysinghe [37] diversified the field further to include multiclass spoofing and behavioral analysis, showing good detection performance but exhibiting constraints because of dataset-based dependency, scaling, and validation of practical deployment. Taken together, the studies indicate the obvious research gap: lightweight, accurate, and generalizable detection frameworks that can work in real-time and across various IoT settings are in demand. Based on these findings, this paper presents a machine learning-based detector that is specially designed to address ARP spoofing MITM attacks, and it aims to achieve a balance between the accuracy of detection and its computational cost, and the ability to adapt to new attack characteristics.

3. ARP Spoofing MITM Detection Model

To successfully detect the ARP Spoofing MITM Attacks in IoT networks, a machine learning-based detection model was developed that incorporates organized preprocessing, supervised classification, and overall evaluation metrics. The model used the IoTID20 dataset. As shown in Fig. 1.

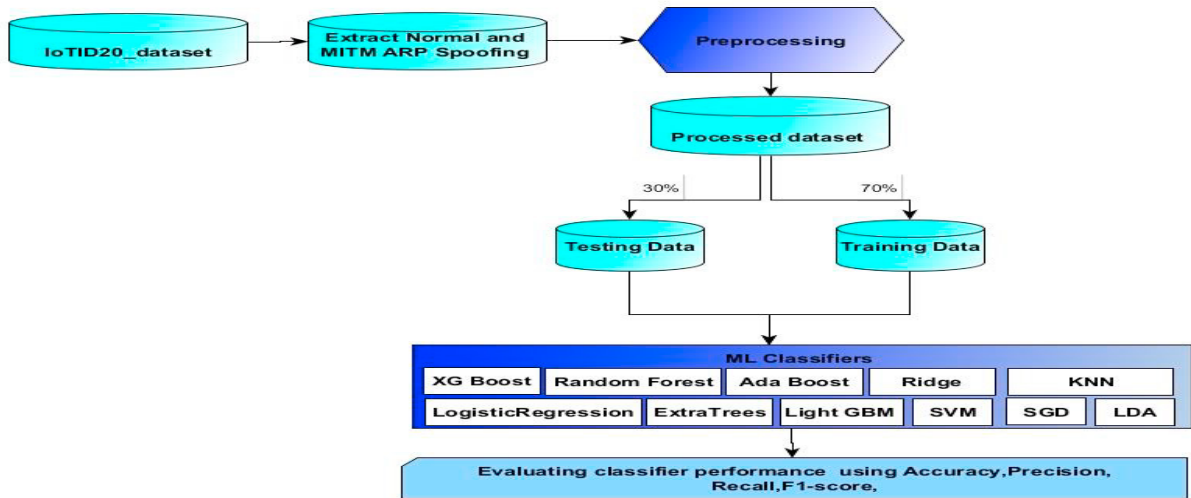


Fig.1: ARP Spoofing MITM Detection model

3.1. Dataset Preprocessing

The IoTID20 dataset [38] was chosen as it includes realistic IoT traffic produced by a smart home setting, and the attack scenarios include ARP spoofing MITM. At this step, the normal traffic and ARP spoofing attack traffic were only extracted, and thus the model is specifically trained to identify benign and malicious ARP communication.

Raw network traffic can be full of noise, redundant attributes, and values that are inconsistent. Preprocessing involved cleaning, normalization of features, and conversion into machine learning formats to prepare the dataset to allow analysis. This made sure that the input features were scaled, balanced, and structured to reduce bias in training [39]. The processed data was separated into two subsets, one of them 70% to be trained and the rest 30% to be tested. The machine learning classifiers were constructed and optimized using the training set, and then the testing set was set aside to be evaluated on with no bias. This division offers a sound balance on adequate training information and powerful performance evaluation.

3.2. Machine Learning Classifiers

A wide range of ML classifiers is used to represent various learning properties and determine the best classifiers for the detection of ARP spoofing. The classifiers include Tree-based models: random forest [40, 41], extra trees [42], XGBoost [43, 44], LightGBM, and AdaBoost[45], which are characterized by strong stability and the ability to perform ensemble learning. Linear and probabilistic models: Logistic Regression, Ridge Regression, and Linear Discriminant Analysis (LDA), which are interpretable and have a low cost to calculate. instance-based and kernel models, K-Nearest Neighbors (KNN) [46] and Support Vector Machines (SVM) [47], which encode spatial and margin-based differences in data. Optimization-based models: Stochastic Gradient Descent (SGD) that offers large-scale solutions. Such a combination of classifiers enables exploring simple and complex decision boundaries, with the view of fully evaluating detection strategies [48, 49].

3.3. Performance Evaluation

Standard performance measures were used to objectively test the performance of each classifier in detecting ARP spoofing MITM attacks: accuracy, precision, recall, and F1-score. These are defined as follows:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP+FP} \quad (2)$$

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

$$F1 - Score = \frac{2 * Precision * Sensitivity}{Precision + Sensitivity} \quad (4)$$

Lastly, the rationale to consider a diverse set of ML classifiers is due to the variability of the IoT traffic and the varying learning biases installed in ML models. The tree-based ensembles (e.g., RF, XGBoost, LightGBM) can handle nonlinear relationships, feature interactions; the linear models (e.g., Logistic Regression, Ridge, LDA) are faster to infer with constrained IoT devices. Margin-based methods (e.g., SVM) and instance-based methods (e.g., KNN) prove to be effective in boundary-sensitive classifications, and optimization-based approaches (e.g., SGD) are scalable. The comparison of these different models in a similar experimental setting guarantees that the presented framework is not inclined towards a specific algorithm and presents the solution chosen is correct and applicable in the real IoT setting.

4. Results and discussion

To test the ARP Spoofing MITM Attacks model, eleven machine learning classifiers were trained and tested on IoTID20 dataset. Accuracy, precision, recall and F1-score were used to measure their performance. The results have been summarized in Table 1, and the discussion below compares the classifiers on each of these metrics.

Table 1. Results

Classifier	Accuracy	Precision	Recall	F1-score
XGBoost	0.995	0.995	0.995	0.995
RandomForest	0.993	0.993	0.994	0.993
AdaBoost	0.990	0.990	0.990	0.990
LightGBM	0.997	0.997	0.997	0.997
LogisticRegression	0.974	0.974	0.975	0.974
ExtraTrees	0.988	0.988	0.989	0.988
SVM	0.976	0.976	0.977	0.976
SGD	0.974	0.973	0.974	0.974
KNN	0.988	0.988	0.989	0.988
Ridge	0.959	0.959	0.961	0.959
LDA	0.959	0.959	0.961	0.959

Accuracy is the percentage of the correctly classified cases. LightGBM with 99.7% has the best accuracy, followed closely by XGBoost 99.5% and Random Forest 99.3%. These three-based ensemble approaches were very effective in capturing non-linear interactions between features and are therefore highly reliable. AdaBoost 99.0%, Extra Trees 98.8% and KNN 98.8% also fared well, whereas SVM 97.6% and Logistic Regression 97.4% gave slightly lower yet competitive accuracy. Ridge 95.9% and LDA 95.9% gave the lowest levels of accuracy, which means those linear models had difficulties with more complicated patterns in ARP spoofing traffic.

Precision is how many of the predicted attack occurrences were attacks. LightGBM 99.7% had the best precision, followed by XGBoost 99.5% and Random Forest 99.3%. This shows that they can reduce false at a high degree, which is essential to minimize unnecessary alerts in an IoT setting. Another effective option was AdaBoost, KNN, and Extra Trees 98.8%-99.0%), and slightly less effective were SVM 97.6% and Logistic Regression 97.4%. Once again, Ridge and LDA 95.9% registered the lowest precision, indicating that they made more misclassifications than the advanced ensembles.

Recalling measures how well it can detect true attacks (reducing false negatives). LightGBM 99.7% and XGBoost 99.5% once again dominated, which implied that they were highly sensitive to ARP spoofing attack detection. Random Forest 99.4% was also very effective as very few attacks were not detected. AdaBoost 99.0%, KNN 98.9%, and Extra Trees 98.9% were found to be the best in terms of recall, whereas SVM 97.7% could be considered as the best model, Logistic Regression 97.5%, and SGD 97.4% were performing reasonably well. The lowest recall, 96.1%, was obtained with Ridge and LDA, which implied a higher chance of missing cases of attack.

F1-score is a balance between the precision and the recall thus a very important parameter in detecting ARP spoofing. Again, LightGBM scored the highest F1-score 99.7%, and thus, LightGBM offered the best overall trade-off between false positives and false negatives. XGBoost 99.5% and Random Forest 99.3% were closely behind to establish how strong they are. Balanced performance was also reflected in AdaBoost, KNN and Extra Trees (98.8%-

99.0%). SVM 97.5% and Logistic Regression and SGD 97.4% had moderate performance whereas Ridge and LDA 95.9% were trailing behind with less balance during classification.

In general, the results show that ensemble gradient boosting models, especially LightGBM, XGBoost, and random forest achieved higher results in all four-evaluation metrics, which confirms their relevance in embodying the complexity of ARP spoofing detection in IoT networks Figure 2 shows comparison of all classifiers performance . Linear models like Ridge and LDA, by contrast, although computationally economical and working well with resource-restrained devices, provided relatively lower accuracy and are thus not so optimal in high-stakes security applications. KNN and SVM (kernel-based) were also noted to be instance-based techniques that were successful in providing good detection rates, as there were identifiable patterns in ARP spoofing traffic. LightGBM was the most effective classifier that obtained almost perfect accuracy, precision, recall and F1-score, whilst the simpler linear models are still useful in the context where the efficiency of the computation is prioritized over the maximum accuracy of the detection.

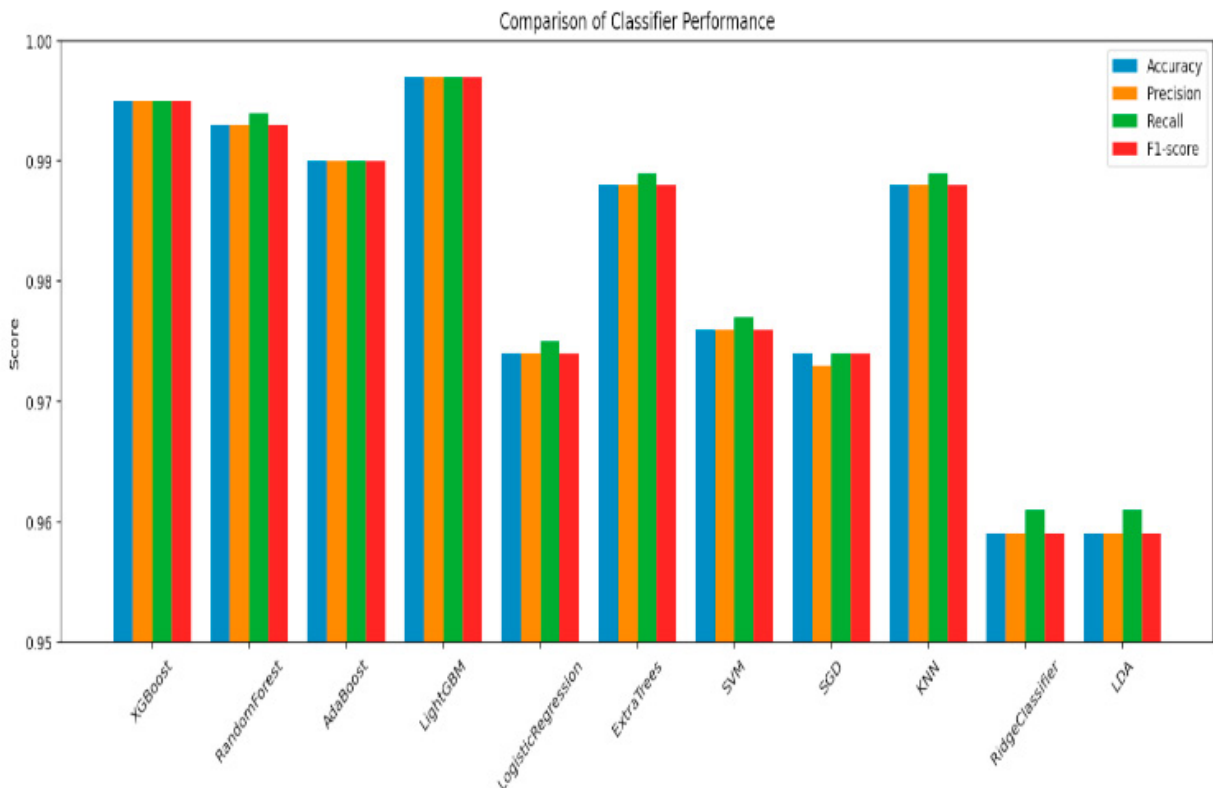


Fig 2. Comparison of all classifier's performance

5. Conclusion and future work

This paper described a machine learning-driven detection system to detect ARP spoofing Man-in-the-Middle (MITM) attacks in an IoT setting. Through the methods of preprocessing the IoTID20 dataset that include systematic preprocessing and the assessment of the broad diversity of classifiers, the study has shown that advanced ensemble models offer a valid and scalable solution to this severe security threat. These findings affirm that gradient boosting techniques, and LightGBM especially, are very accurate and resilient at detecting ARP spoofing and thus are good candidates in the actual implementation of IoT security. Meanwhile, some less accurate linear models, including Logistic Regression, Ridge, and LDA, demonstrate the possibility of lightweight solutions in cases where the performance of peak detection is not the priority and computational efficiency is. Despite these additions, the research is limited by the fact that it uses one dataset and is tested in an offline setting. Practical IoT applications might pose more issues like the heterogeneity of traffic, the scarcity of resources, and dynamic opponents, who can circumvent fixed detection models. The solution to these dilemmas is the foundation of future employment. Future developments are facilitating the framework validation on larger and more recent data sets, incorporating deep learning models or

hybrid ML-cryptographic models to achieve increased robustness, and building lightweight real-time systems that can be deployed to more limited IoT devices. Moreover, adversarial robustness and online learning methods are to be investigated to improve the flexibility to the varied ARP spoofing techniques.

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